AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX

MODEL EXAMINATION PAPER 2026 AND ONWARDS

Chemistry Paper II

Time: 1 hour 50 minutes Marks: 25

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Paper 2026
Learning Only

INSTRUCTIONS:

1. Check your name and school information above. Sign if the information is correct.

I agree that this is my name and school. **Candidate's Signature**

2. Read each question carefully.
3. There are SIX questions in this exam. Answer ALL questions. Questions 5 and 6 each offer TWO choices. For each of these questions, attempt ONE choice only.
4. When answering the questions:
   - Use a black pen/ pointer to write your answers. DO NOT write your answers in pencil.
   - Use a black pencil for diagrams and drawings. DO NOT use coloured pencils, markers or highlighters.
   - DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
     If you make a mistake, cross it out neatly and write the correct answer beside it.
   - Write your answers only in the allocated space provided for each question.
   - DO NOT write outside the answer space. Answers written outside the allocated space will not be marked.
5. The marks for each question are shown in brackets ( ) next to the question number. Use this information to manage your time efficiently.
6. DO NOT mention your name or school name anywhere inside the paper.
7. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Determine the chemical formula for carbon disulphide by completing the given table.

(**Note**: ${}^{12}_{6}\text{C}$ and ${}^{32}_{16}\text{S}$)

Compound	Valency of Carbon	Valency of Sulphur	Chemical Formula
Carbon disulphide			

Q.2. (Total 3 Marks)

An atom of an element **X** has atomic number 12 and mass number 24. It loses 2 electrons from its outer most shell and acquires a +2 charge.

Draw the atomic structure of the ion of element **X**.

Space for drawing

Q.3. (Total 4 Marks)

Explain the electronegativity trend across the period and down the group in the periodic table.

Across the period: _____

_____

_____

_____

Down the group: _____

_____

_____

_____

Q.4. (Total 3 Marks)

Consider a closed container with liquid having weak intermolecular forces. This liquid experiences both, vapour pressure and boiling point at constant temperature.

Assess the influence of weak intermolecular forces on the vapour pressure and the boiling point of the liquid. Support your answer with scientific reasoning.

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Q.5. (Total 6 Marks)

EITHER

a.

i. Balance the given chemical equation by inserting numbers where necessary. (1 Mark)

$$____ CH_{4(g)} + ____ O_{2(g)} \rightarrow ____ CO_{2(g)} + ____ H_2O_{(l)}$$

ii. Calculate the mass of carbon dioxide and water produced on complete combustion of 0.25 mol of methane.

(**Note**: Molar mass of C = 12 g, O = 16 g and H = 1 g. To solve the numerical, use equation in part i after balancing.) (5 Marks)

OR

b. Given below are four elements.

Non-Metal	Metal
Chlorine $\left({}^{35.5}_{17}\mathrm{Cl}\right)$	Magnesium $\left({}^{24}_{12}\mathrm{Mg}\right)$
Oxygen $\left({}^{16}_{8}\mathrm{O}\right)$	Potassium $\left({}^{39}_{19}\mathrm{K}\right)$

i. Identify TWO elements from the given table which, when form ions, show the same electronic arrangement as a neon $\left({}^{20}_{10}\mathrm{Ne}\right)$ atom. (2 Marks)

ii. Illustrate the formation of an ionic compound using the elements identified in part i with the help of dot and cross structure of the compound. (4 Marks)

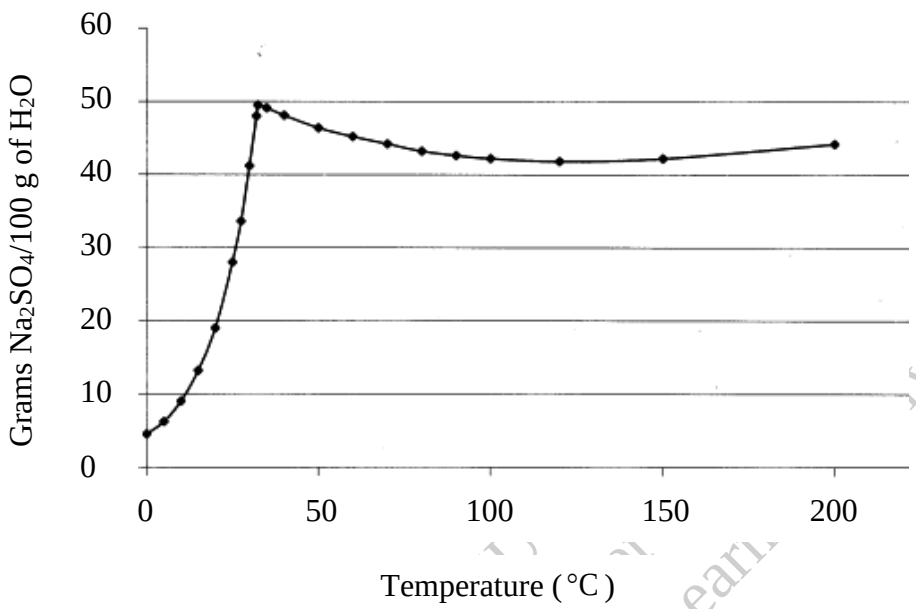
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Q.6. (Total 6 Marks)

EITHER

a. The given graph represents the effect of temperature on the solubility of sodium sulphate (Na_2SO_4).



NOT TO SCALE

Assess the graph to answer the following questions.

i. Identify the highest and the lowest solubility values of Na_2SO_4. (2 Marks)

ii. Evaluate the solubility trend of Na_2SO_4 with temperature. How does it differ from the general solubility rule and why? (4 Marks)

OR

b. A student sets up an electrolytic cell using molten potassium chloride (KCl). Electrodes are connected to a power source and electrolysis begins.

i. Deduce the direction of movement of cations and anions during electrolysis. Construct balanced chemical equation to represent the reactions occurring at each electrode. (4 Marks)

ii. Evaluate the role of the power source in the electrolytic process and predict what would happen if the power source was removed. (2 Marks)

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END OF PAPER

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